Homework 1

Prepare your answers as a **single PDF file**.
Group work: You may work in groups of 1-3. **Include all group member names in the PDF file.** Only one person in the group should submit to Canvas.
Due: check on Canvas.

1. Upload a photo of your group. Please put your names on the picture too. (You must answer this question even if you work by yourself).

2. Suppose you are managing an animal shelter and you have a dataset of animals that were in the shelter and how long they waited before being adopted. For instance, like this:

Name	Species	Age (years)	Weight (lbs)	Days before adopted
Fido	Dog	0.75	15	15
Rover	Dog	2.5	22	108
Felix	Cat	1.5	12	52
…	…	…	…	…

(a) Your goal is to use this dataset to predict how long a new animal will likely stay in the shelter. What type of machine learning problem is this? (Pick one of the following)

a. Classification: Supervised learning where the mode predicts a *discrete category/ label*. The output is a class.

b. Regression: Supervised learning where the model predicts a *continuous numeric value*. The output is a number

c. Clustering: Unsupervised learning that groups similar data points together *without any labels.* The algorithm discovers structure on its own.

d. Semi-supervised learning: A mix of supervised and unsupervised; uses a *small amount of labeled data* alongside a large amount of unlabeled data to improve learning.

e. Self-supervised learning: The model generated its *own labels from unlabeled data itself*. No human input for labeling is required.

f. Reinforcement learning: An agent learns optimal actions by *trial-and-error,* receiving rewards or penalties from its environment.

(b) List the numeric variables and the categorical variables:

- **Numeric variables:** Age (years), Weight (lbs) , Days before adopted
- **Categorical variables:** Name, Species

3. What is the purpose of calculating correlations during exploratory data analysis? Answer in 2-3 sentences.

Calculating correlations during exploratory data analysis helps identify which input features have a strong linear relationship with the target variable, so you can focus on the most predictive features. A pitfall that is mitigated is **multicollinearity** when 2 input features are too correlated.

4. What is meant by “one-hot encoding”? If one-hot encoding was performed on the table shown in Q2, how many new rows and columns would be created? Answer in 2-3 sentences.

One-hot converts categorical variables into multiple binary (0/1) columns. This helps ML algorithms read it properly and does not show an ordinal relations. Assigning numbers Dog 1, Cat 2, Bird 3 etc implies false ordinal relations.

5. The textbook discusses three approaches for dealing with missing data values. Describe *any one* approach in 2-3 sentences.

There are three approaches: **removing rows**, **removing columns and imputation**. Removing rows and columns can lead to data loss.

Imputation: replaces missing values with calculated estimates typically with the median, but you risk fudging not real data. Which we discussed in class

6. What is meant by “Min-Max scaling”? Answer in 2-3 sentences. What would be the output of applying Min-Max scaling to this dataset: (to be done by hand)

Min-Max scaling (also called *normalization*) rescales a numeric feature so all its values fall within a fixed range, usually 0 to 1. The formula is as follows:

$x' = (x - x_{min}) / (x_{max} - x_{min})$

Applying Formula

Preg	**BP**
2 -> (2-1)/2 = **0.5**	74 -> (74-54)/20 = **1.0**
3 -> (3-1)/2 = **1.0**	58 -> (58-54)/20 = **0.2**
2 -> **0.5**	58 -> 0.2
1-> **0**	54 -> 0
2 -> **0.5**	70 -> (70-54)/20 = **0.8**

Preg Scaled	**BP Scaled**
0.5	1.0
1.0	0.2
0.5	0.2

0	0
0.5	0.8

Preg: min=1, max=3 BP: min=54, max=74

7. Write Python code to develop and evaluate a regression model to predict median house value for the housing dataset in the textbook. Specifically, do **only** the following:

1. Download the housing dataset used in the textbook code
2. What is the target variable? _median_house_value_
3. Plot a scatter plot of median_income and median_house_value [show plot]
4. Remove all rows where median_house_value >= 500,000
5. Split data into test and train subsets with 20% in the test subset using random sampling
6. Create a new variable in the train subset that is total_bedrooms/total_rooms.Calculate the correlation coefficient of this new variable with median_house_value. [show correlation coefficient]
7. Create a pre-processing pipeline that does the following:
    a. For the categorical variable ocean_proximity, replace missing values with the most frequent value, followed by one-hot encoding
    b. For the numerical variables "housing_median_age", "total_rooms", "total_bedrooms", "population", "households", and "median_income", replace missing values with the median, followed by a log transform, and then scaling to a 0-1 range.
    c. Create a new variable defined as total_bedrooms/total_rooms, then scaling to a 0-1 range.
    d. How many variables are in the pre-processed training dataset?
8. Add a LinearRegression model to the preprocessing pipeline. Use 3-fold cross validation to calculate the RMSE error on the training dataset. [Show RMSE error]
9. What is the RMSE error on the test dataset?

    Hint: follow the steps in the 02_end_to_end_machine_learning_project.ipynb notebook. Include **only** the relevant portions in your code.
    Submit your code as a clickable link to a Google Colab notebook.

LINK: Question 7 colab link